Language Features under Weight Quantization: Activation Agreement and Intervention Effects

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Abstract

Quantization makes language models cheaper to run, but it can affect languages unequally. Interpretability work explains multilingual behavior through sparse-autoencoder (SAE) features that fire for specific languages and can be edited (Deng et al., 2025), and recent work asks whether these features keep their activations after quantization (Duan, 2026). Whether they also keep their function is untested. We therefore ask: after a model's weights are rounded to fewer bits, do language-selective SAE features still do the same job? Reproducing the activation study first showed that omitting the beginning-of-sequence token distorts its Gemma results. With corrected inputs, we remove, scale and restore each language's feature group in Gemma-2-2B across 27 FLORES-200 languages and 8- to 4-bit weights, compare against matched controls, and repeat the main tests on Gemma-2-9B. Removing a language's features harms its next-token prediction in all 27 languages, and harm rises smoothly with edit strength; it stays near full precision down to five bits and grows at four. Yet activation and function come apart: at four bits only 13.2% of features pass the activation-survival cutoff, while 99.7% are still identified by their output effect. Control directions drift almost equally, and restoring the features repairs only 2.6% of the loss. Activation agreement and the effect of an edit therefore need separate measurements. Code and results are at https://github.com/TH4KUR/always-late.

1 Introduction

Quantization rounds model weights to fewer allowed values. It can make language models cheaper to store and run, but may also change their behavior. The effects can differ across languages and tasks (Marchisio et al., 2024). We investigate what happens to the internal features used to explain these models.

A sparse autoencoder (SAE) is a separate network that describes a model's hidden state using a few active features from a large learned dictionary. Each feature has an activation strength and a direction in the hidden state. Some activate more often for one language than for others. Previous work has identified and edited such language features (Deng et al., 2025); another study measures how their activations change under quantization (Duan, 2026).

We connect these measurements and find a sharp dissociation. At four bits under simple rounding, only 13% of language features pass the activation-survival cutoff, yet 99.7% can still be identified from what they do to the model's output. Activation agreement, task performance and preservation of effect need separate measurements: they answer different questions and give different answers. A changed activation does not tell us whether removing that feature still has the same effect. A feature can matter for predicting the next token without producing a measurable change on a reading-comprehension test.

Four questions organize the completed study:

RQ1: Which language features retain their activation patterns after quantization?

RQ2: How does editing language features affect text prediction and comprehension before and after quantization?

RQ3: How well do activation measurements predict whether a feature's output effect is preserved?

RQ4: How much does the same feature edit change when the model is quantized, and what explains that change?

These final questions organize the accumulated evidence. They do not replace the original hypotheses or the choices fixed before each analysis.

The mechanistic element is the use of SAE features as causal units. Rather than measuring only whether quantization changes model outputs, we remove, scale and patch individual features and measure whether the same intervention still produces the same downstream effect. This tests where in the model quantization matters and whether the internal explanation survives compression.

2 Related Work

Quantization affects languages unequally. Quantization studies usually judge a compressed model by its outputs. Marchisio et al. (2024) find that quantization harms some languages more than others. This raises our question: what changes *inside* the model for each language? To group languages we use the resource classes of Joshi et al. (2020). These describe available data, not what a model saw, so tier differences are only descriptive. We compare simple rounding with GPTQ, which uses calibration data to correct rounding error (Frantar et al., 2022).

SAEs reveal language-specific features. Sparse autoencoders rewrite a hidden state as a few active features from a learned dictionary (Gao et al., 2024; Rajamanoharan et al., 2024). Gemma Scope releases such SAEs for Gemma 2 (Lieberum et al., 2024), giving one fixed dictionary before and after quantization. With SAEs, Deng et al. (2025) found features specific to one language and showed that editing them changes the model's output in that language. These features are our starting point: we ask whether their causal role survives quantization. But SAE features are not canonical units and can merge or split concepts (Leask et al., 2025), and good proxy scores need not mean useful features (Karvonen et al., 2025). Following Makelov et al. (2025), we therefore judge features by the effects of controlled edits, not by activation agreement alone.

Do features survive compression? Closest to our question, Duan (2026) measure whether SAE feature activations still match after quantization and argue that perplexity can miss this damage; we reproduce and extend their study. Borobia et al. (2026) ask a similar question for pruning. Neither tests whether a feature still *does the same job*. Hoang et al. (2026) show that a feature's downstream effect can be studied in its own right, which supports comparing effects directly. To locate where an effect changes we use activation patching, as Chang et al. (2025) do for quantization errors, and report an interaction term because patches can interact (Vaidyanathan et al., 2026). For a task beyond next-token prediction we use the Belebele reading-comprehension benchmark (Bandarkar et al., 2024).

3 Experimental Setup

Models and data. The main model is Gemma-2-2B (Gemma Team et al., 2024). Its fixed Gemma Scope JumpReLU SAE has 16,384 features and reads the hidden state after layer 12, near the midpoint of 26 transformer blocks. Prior work does not point to a single best layer. Language-specific neurons sit mainly in the first and last layers (Tang et al., 2024; Kojima et al., 2024), middle layers work in a more shared, English-leaning concept space (Wendler et al., 2024), and language-specific SAE features appear across Gemma 2 layers with layer-dependent effects (Deng et al., 2025). If language-specific units concentrate at the extremes, finding language-selective features and causal effects at a middle layer is a conservative test, and early or late layers might show stronger effects. We study one middle layer with one public SAE because each added layer or dictionary repeats feature selection, matched controls and every quantized run, which our compute budget did not allow. Model inference uses bfloat16; SAE calculations and edits use float32. The Gemma-2-9B extension uses a layer-20, 16k SAE at a similar relative depth in a 42-layer model. A smaller Llama-3.2-1B study (Grattafiori et al., 2024) uses a layer-12 TopK SAE with 14,336 features and $k = 48$. Model, SAE, training data and tokenizer change together in that comparison.

The main panel contains aligned FLORES-200 sentences (NLLB Team et al., 2022) in 27 languages, nine per descriptive resource tier. Feature selection uses the development split. Activation and language-association tests use 27,036 content tokens per language from devtest; prediction-loss tests use 300 aligned devtest sentences. The comprehension test uses 385 Belebele questions per language in 18 languages that passed calibration, grouped into 219 passage clusters. A second transfer panel uses fresh Wikipedia articles in six languages.

Quantization. Round-to-nearest (RTN) maps a weight to its nearest allowed value. We use three quantizers: legacy RTN, the setting of the study we

reproduce, which shares attention scales across corresponding coordinates of different heads; matched RTN, which uses one scale per output channel; and GPTQ, calibrated on 8,192 Wikipedia tokens per language that are separate from evaluated text. Main figures show legacy RTN for direct comparison with the reproduced study; the other two quantizers give the same qualitative pattern (Appendix C).

Selected transformer weights are rounded; embeddings and normalization parameters stay unchanged. Values remain stored and evaluated in floating point, so we measure weight-rounding effects rather than deployment speed or memory savings. The original tests cover FP and eight to four bits (INT8–INT4). FP is the unquantized reference. Later intervention comparisons focus on INT6 and INT4. INT16 and INT12 make small real weight changes and serve as reference conditions; INT16 is not identical FP arithmetic.

Analysis status. Most FLORES studies are exploratory: they reuse examined sentences or follow earlier results. Fixing a later stage's choices before its output does not supply independent confirmation. The original held-out Belebele family is confirmatory. We claim no verified external preregistration. This reporting revision adds no experiment; the feature-count comparison in Section 6 is a new descriptive summary of saved outputs.

4 Problems Found and Corrected

The missing BOS token in the earlier Gemma evaluation. Our first reproduction disagreed with Duan's reported survival rates. Running their released code reproduced their table closely: INT7 survival was 81.19% versus 81.26%, and INT6 was 51.15% versus 51.30%. We traced the large disagreement to a methodological problem in their Gemma input protocol: it omitted the beginning-of-sequence (BOS) token.

Adding BOS to the same 256-content-token WikiText windows, while excluding its position from feature statistics, reduced full-precision perplexity from 458.4 to 16.1. Lower perplexity means better text prediction. INT6 survival rose to 97.52%, and INT7 to 99.97%. Each regime defines its own active-feature set, so these percentages do not follow exactly the same feature population. We corrected our subsequent protocol by prepending one BOS token and excluding BOS and padding from statistics and edits. This demonstrates a large input-protocol effect in the Gemma evaluation; it does not establish that every conclusion in Duan's paper is wrong.

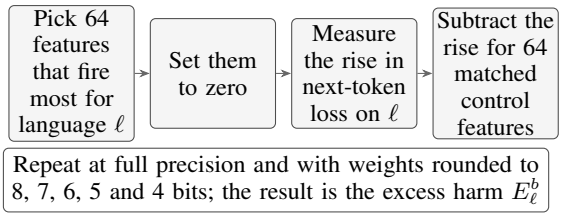


Figure 1: The core test, run for each of the 27 languages.

Our own corrections. We retired the first pipeline after finding separate flaws, including counting BOS in feature statistics and changing tokens during decode/re-encode. Its results are excluded. We also corrected an assumption that INT16 was identical to FP: re-rounding makes a small real weight change. A small gain experiment had to be rerun on the intended GPU because the hardware shift was comparable to the measured effect. The 9B review found two defective prediction criteria, described with its results below.

5 Measurements and Controls

Figure 1 shows the core test: remove a language's most selective features, measure how much worse the model predicts that language's text, and compare with removing matched ordinary features.

Feature activation and preferred language. A feature fires when its activation is positive. We select language features using the ratio of their highest per-language firing rate to their mean across languages. Selection also requires enough active tokens, a shuffled-language cutoff (50 permutations), and stable peak language in 80 of 100 resamples. This yields 8,086 features. On FP devtest, 6,663 have a unique supported peak language. We test retention of that language only for this established set.

Activation correlation instead compares the same feature's values on the same tokens before and after quantization:

$$R_1 = \mathrm{corr}_t(a_f^{\mathrm{FP}}(t), a_f^b(t)). \quad (1)$$

The historical survival rule is $R_1 > 0.9$. Retaining a preferred language and retaining high correlation answer different questions.

Edits and prediction loss. Each language group contains its 64 selective features with the highest

mean development-set activation, called K64. Control groups come from 2,860 supported nonselective features and match firing rate and mean activation on a log scale. Evaluation uses three control draws; the saved manifest stores up to five. Matching is close but imperfect: 75 of 81 language–size ratios fall within $[0.8, 1.25]$, with Thai, Yoruba and Lao outside at K64.

For hidden state x, activation a_f and decoder direction d_f, we edit a group G as

$$x' = x + \sum_{f \in G} (a'_f - a_f) d_f. \tag{2}$$

This retains the SAE reconstruction error. Removal sets $a'_f = 0$, scaling sets $a'_f = \alpha a_f$, and a donor patch uses the FP activation at the same position. For example, Turkish has 64 selective features; setting all 64 to zero and measuring the increase in next-token loss for Turkish sentences, minus the increase from zeroing a matched non-selective group, gives the excess harm E.

Prediction loss, or negative log-likelihood (NLL), measures how much probability the model assigns to the actual next token. Lower is better; units are nats/token. We subtract control-group harm from language-group harm and call the result excess harm, E^b_ℓ. Positive E means the language edit hurts more. The effect ratio $E^b_\ell / E^{\mathrm{FP}}_\ell$ compares it with FP: one means equal harm, 1.5 means 50% greater harm. The median of language ratios, their mean, and the ratio of mean effects are reported separately because small denominators can make them differ. Patch recovery measures how much of the loss gap a specified restoration removes.

Output-effect similarity. A feature's effect fingerprint is the vector of changes in output-token scores caused by removing it. Cosine similarity with its FP fingerprint, W_1, measures how closely those directions agree; one means identical direction. The F5 study samples 1,487 features; 1,412 pass the INT16 reliability criterion. A stricter similarity cutoff, 0.9489, comes from the fifth percentile of INT16 cosines. Close agreement with this reference and recognizable feature identity are different tests.

Bracketed ranges are 95% bootstrap intervals unless stated otherwise (resampling units in Appendix A).

6 RQ1: Which Activations Survive?

Selective features are not more fragile. Language-selective features survive only slightly less often than matched controls: 0.28 points less at INT7 and 1.44 at INT6, not significant after correction for multiple comparisons (adjusted $p = 0.514$ and 0.128). The non-significant tests are these selective-versus-control survival contrasts; the study's main effects (excess harm and dose–response in 27 of 27 languages, 99.7% effect identity, 2.6% repair) have intervals far from zero. The small gap is explained by activation strength rather than low power. Selective features activate more weakly than controls (median peak activation 9.8 versus 14.2 at the stricter cutoff $S > 5$), and weaker features survive less (Spearman $\rho_s = 0.35$ at INT6). At $S > 5$ the gap widens to -7.16 points, but when controls are also matched on activation strength it reverses to $+4.58$. Activation strength, not language selectivity, predicts which features stop firing.

Low-, mid- and high-resource languages have INT6 survival rates of 69.4%, 90.6% and 95.1% (Appendix Figure 6). Accounting for activation scale reduces the fitted tier slope from $+12.3$ to -0.81 points. Later measures still show some tier patterns at INT16, and GPTQ's adjusted result depends on the adjustment choice. These measurements alone do not identify a training-resource mechanism behind language damage.

Resource tier also does not predict how many language-selective features a language has (Appendix Figure 6b; descriptive, post hoc). Counts range from 73 (Portuguese) to 1,719 (English), and Lao, Yoruba and Khmer each have more than every high-resource language except English, Japanese and Chinese (Spearman $\rho_s = -0.07$ between count and tier). Selectivity is relative to the other 26 languages: a feature counts when it fires for one language much more than for the rest. A distinctive script or spelling can give a language many such exclusive features (Lao 464, Khmer 444, Yoruba 453), but script alone does not decide it: Bengali and Telugu have their own scripts and few features. The lowest counts belong to languages with a close relative in the panel, such as Portuguese and Spanish or Russian (83) and Ukrainian (84), whose shared features are exclusive to neither; English, the dominant training language, is the outlier. Feature count therefore reflects how distinct a language's input looks to the model relative to the panel, not how

well-resourced it is. Count is also unrelated to survival ($\rho_s = -0.16$ at INT6, -0.07 at INT4), and every removal test uses 64 features, so count differences do not drive the harm comparisons.

Other quantizers. Under matched RTN and GPTQ, more features survive and loss rises less, but the pattern is the same: survival falls with fewer bits, and no consistent selective penalty appears across the three methods (Appendix Table 2).

Correlation misses some activation changes. Multiplying every activation by the same factor can leave correlation unchanged. We therefore decompose quantized activation as a scaled FP signal plus a residual: $b = ga + c + e$. Here g is gain, c is an offset, and e is the part scaling does not explain. The residual is uncorrelated with a. Where the quantities are defined,

$$\lambda = \log(R_1^{-2} - 1) = \log \sigma_e^2 - 2 \log |g| - 2 \log \sigma_a. \quad (3)$$

The correlation cutoff 0.9 corresponds to $\lambda < -1.450$, with positive gain. We also measure preactivation change divided by the feature's learned threshold, denoted ν, on tokens active in FP. This puts the change in threshold units.

The audit covers 214,506 active language–feature pairs. Across six INT4/6 conditions, rank correlations with FP activation spread are -0.507 to -0.534 for λ, versus -0.004 to 0.114 for ν. Threshold scaling reduces this dependence, though dependence on peak activation remains. These measures explain different parts of activation change; the prediction test below asks whether they add useful information.

7 RQ2: What Do Language-Feature Edits Change?

Removing features harms next-token prediction. FP excess harm is positive in all 27 languages of Gemma-2-2B (Figure 2a): mean 0.704 nats/token $[0.684, 0.723]$, median 0.262. The effect stays near FP at INT6, then grows at INT4 (Figure 2b). Legacy INT4 has a median ratio of 1.54 $[1.43, 1.62]$, a mean ratio of 2.14 $[2.04, 2.50]$, and a ratio of mean effects of 1.33. The same increase appears, at a smaller size, under matched RTN and GPTQ. Greater removal harm indicates greater sensitivity to the edit.

Gradual suppression supports the same result (Figure 3). In all 27 languages, FP harm increases as the activation multiplier decreases. Median harm is 0.0223, 0.0902, 0.1687 and 0.2624 at multipliers 0.354, 0.177, 0.088 and zero. The strict rule requiring monotonic change across every precision fails because of a small intermediate-width dip; the suppression-dose and consistency rules hold.

A second Gemma size and a different setup. Gemma-2-9B shows positive FP excess harm in all 27 languages, with median 0.270 $[0.197, 0.608]$. The median effect ratio is 1.019 $[0.978, 1.045]$ at INT6 and 1.586 $[1.238, 2.264]$ at INT4. At INT4, the mean ratio is 2.479 and the ratio of mean effects is 1.474. These intervals resample languages. Paired 2B–9B difference intervals include zero, so the data support the broad pattern across sizes without resolving a size effect.

Two of the predictions fixed before the 9B run were badly specified: one stated the suppression dose–response in the wrong direction, and one could not fail given how features were selected. We do not count them as confirmations. The measured suppression-harm ordering holds in all 27 languages. The 9B run also covered matched RTN and GPTQ. The remaining frozen predictions for both tiers were met.

Llama has a smaller positive FP median effect, 0.048 $[0.018, 0.173]$, with positive effects in 21 of 27 languages. This misses the consistency rule requiring 24. Its INT4 median ratio is 0.86 $[0.74, 1.03]$, with less consistent suppression ordering. The effect differs across these model/SAE setups.

The comprehension effect is not established. On held-out Belebele, FP excess correct-option loss is -0.0001 nats/item $[-0.0035, +0.0031]$. Accuracy is about 42.6% unedited, 42.6% after language-group removal and 42.5% after control removal. The required positive FP effect is absent, so the five planned conditional retention statements cannot be interpreted. Larger nonselective edits (1,024–2,860 features) lower accuracy by 1.2–2.2 points; other-language selective edits reach 6.0 points. Sham edits change nothing. GPTQ's INT4 baseline accuracy is 31.8%, above 25% chance, and the K64 effect remains small. The test detects larger edits but leaves small K64 effects possible.

Repair gains are limited. Restoring FP activations in K64 (Figure 4c) recovers 2.6% $[2.4, 2.8]$ of the legacy INT4 loss gap, versus 39.3% $[38.5, 40.0]$ for the full residual. GPTQ gives a similar 2.0%. These are measured patch benefits, not additive

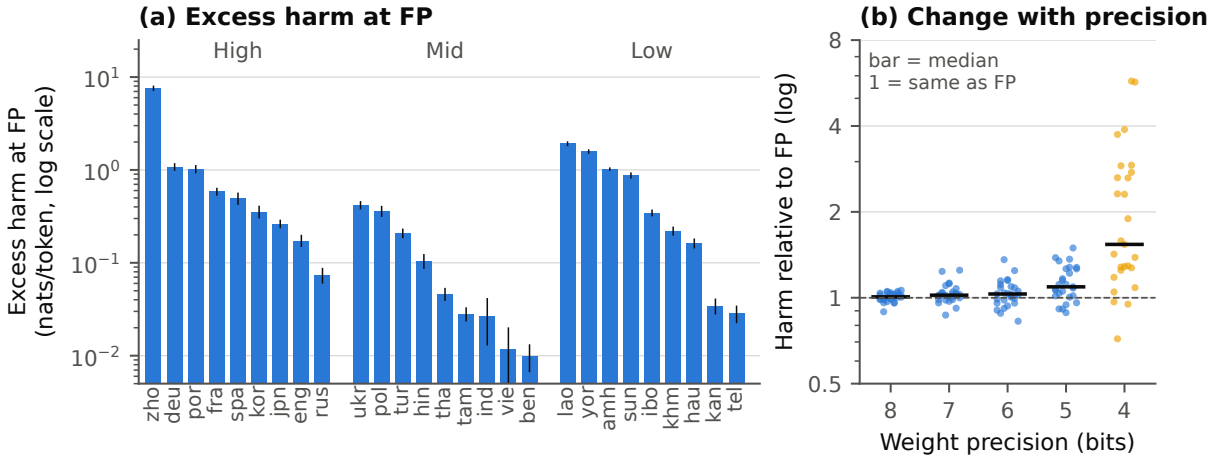


Figure 2: Removing a language's 64 selective features (K64) harms that language's next-token prediction (Gemma-2-2B, 300 FLORES sentences per language). (a) Excess harm at FP, language-group harm minus matched-control harm, with 95% bootstrap intervals; it is positive in all 27 languages. (b) Each dot is one language's excess harm at a lower precision divided by its FP value; bars mark the median. The effect stays close to FP down to five bits and grows at four bits (median 1.54).

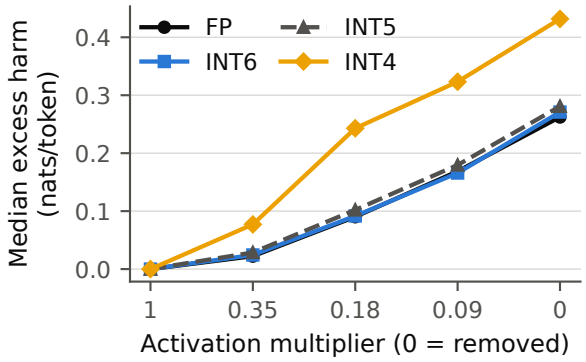


Figure 3: Graded suppression (Gemma-2-2B, legacy RTN). Multiplying a language group's activations by a smaller factor causes more harm; 0 is full removal. Lines are medians over 27 languages. FP, INT6 and INT5 overlap; INT4 is higher at every step.

shares of the cause of model damage.

The main fine-gain study fails its repair rules. A legacy-RTN follow-up on 562 fresh sentences finds a small benefit at multiplier 1.1: -0.00135 nats/token, with 97.5% interval $[-0.00197, -0.00032]$, and improvement in 20 of 27 languages. It meets the rule after multiple-comparison correction at 1.1, but not 1.2. The primary pair of multipliers uses Holm correction. GPTQ does not repeat the fine-gain trend on the main 300-sentence panel; the fresh-sentence test was legacy-only.

Selecting groups anew after quantization changes their membership (overlap with FP falls from Jaccard 0.855 at INT6 to 0.488 at INT4), but does not make removal more harmful (median difference -0.0098). The most effective activation multiplier is repeatable within the study, yet does not track inverse activation shrinkage (Spearman $\rho_s = -0.17$, $p = 0.804$). Simple shrinkage compensation is therefore not supported by this test.

8 RQ3: Do Activation Measures Predict Output Effects?

Of the 8,086 selected features, 6,663 establish a peak language on FP devtest. At legacy INT4, 70.8% retain it, while only 13.75% in this group pass activation correlation 0.9. Near-tied FP language preferences are more likely to switch: peak-margin discrimination scores are 0.90, 0.81 and 0.75 at INT6, INT5 and INT4. Removing matched lost-language and retained-language subgroups gives little harm difference: medians $+0.0006, +0.0007, +0.0002$ nats/token at INT4/5/6.

The individual-effect experiment shows that activation change can coexist with recognizable function (Figure 4a). In the same 1,487-feature sample, only 13.2% pass the activation cutoff at legacy INT4, but median output-effect cosine is 0.839 $[0.829, 0.848]$. Most effects remain distinctive: 99.9% exceed different-feature controls and 99.7% identify their own FP feature. Only 0.3% meet the stricter INT16-based similarity cutoff. Keeping the same language label does not reliably identify which effects meet the preservation criterion; discrimination scores are near chance (legacy 0.506, GPTQ about 0.49).

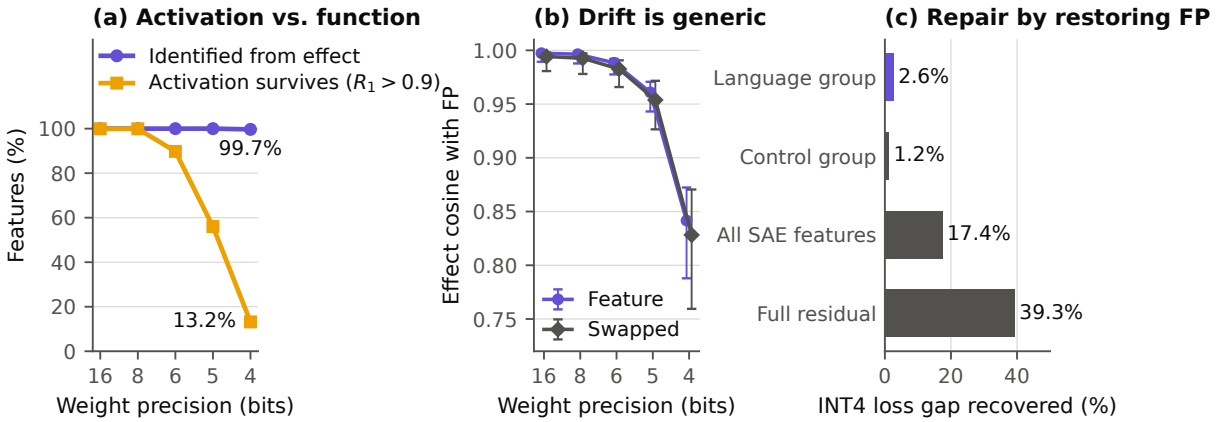


Figure 4: Activation survival and functional preservation come apart (Gemma-2-2B). (a) In the 1,487-feature F5 sample under legacy RTN, the share passing the activation-survival cutoff falls to 13.2% at INT4, while 99.7% of features are still identified as their own FP feature from their output effect. (b) Median output-effect cosine with FP (interquartile range) for 404 features, removing the feature's own direction or a swapped direction at the same positions and strengths: both drift almost equally, so the INT4 drift is mostly generic. (c) Share of the INT4 loss gap recovered by restoring FP activations: the language group recovers 2.6%, about as little as a matched control group and far less than the full residual.

Continuous activation correlation is more informative. On 1,412 reliable features, we predict effect similarity while holding out one language at a time. Adding correlation to activity, context-count and FP-effect-size predictors reduces mean absolute error by 3.1–4.4% across all six quantizer/width conditions. Every language-bootstrap interval is above zero. At INT4, the gain is 3.7% $[2.3, 5.2]$ for legacy RTN and 3.5% $[2.1, 5.0]$ for GPTQ. These are small positive gains below the project's 10% practical target. Partial rank correlations are 0.239–0.399.

Adding gain and threshold-scaled change ν after correlation gives further improvements of -0.4 to $+1.8$%, with every interval including zero. The measures help describe activation changes without showing an additional prediction benefit. Llama's 610 reliable features show a similar INT4 gain, 4.2% $[1.6, 6.7]$, with partial correlation 0.252 $[0.184, 0.322]$; INT6 is uncertain.

9 RQ4: Does the Same Edit Still Have the Same Effect?

Separating earlier context from later weights. We fix the edit direction and strength at their FP values. We then cross FP and quantized earlier layers with FP and quantized later layers. Let $D[p, s]$ be the resulting output-logit change, subtracting each condition's own unedited output. With Q denoting quantized,

$$C = D[\mathrm{Q}, \mathrm{FP}] - D[\mathrm{FP}, \mathrm{FP}], \quad (4)$$

$$W = D[\mathrm{FP}, \mathrm{Q}] - D[\mathrm{FP}, \mathrm{FP}], \quad (5)$$

$$I = D[\mathrm{Q}, \mathrm{Q}] - D[\mathrm{Q}, \mathrm{FP}] - D[\mathrm{FP}, \mathrm{Q}] + D[\mathrm{FP}, \mathrm{FP}]. \quad (6)$$

C measures the context contribution, W the later-weight contribution and I their interaction. The vectors add exactly; their lengths and medians do not. For groups, we compute the same crossed contrasts directly in prediction loss. A fifth condition uses the quantized model's own activation strength.

The panel has 404 features and 6,464 feature–context pairs. We retain 2,203 pairs from 235 features whose FP effects exceed five INT16 reference changes. A matched different-feature direction and three random directions with equal norm test sensitivity shared with other edits. Prediction holds out feature identities, while source sentences may occur in both splits.

At legacy INT4, median group excess harm is 0.848 nats/token for the fixed-strength edit, versus 0.263 in FP and 0.433 using native quantized strength. Median absolute context and weight terms are 0.310 and 0.104 nats/token (Figure 5c). Native-minus-fixed harm is negative in every language, with median -0.351 $[-0.358, -0.293]$. The native-minus-weights-only remainder is smaller in the median language ($+0.0138$ $[0.0093, 0.0233]$), with variation across languages. For individual output vectors, the

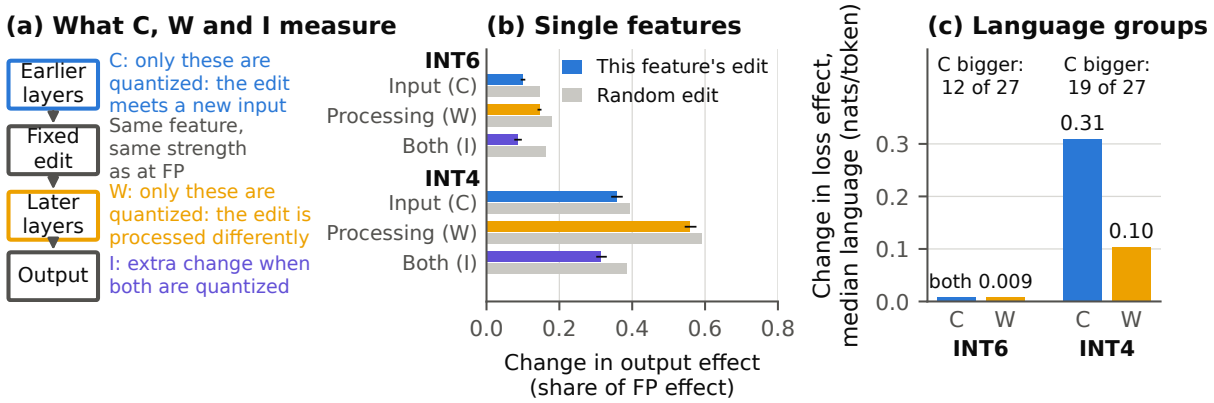


Figure 5: Where the change in a fixed feature edit comes from (Gemma-2-2B, legacy RTN). (a) The edit stays as at FP; rounding only the layers before it gives C (a new input), only the layers after it gives W (different processing), and the extra change when both are rounded gives I. (b) Single features: median size of each part as a share of the edit's FP effect (coloured, 95% intervals), beside the same for random edit directions (grey), which change about as much. (c) Language groups: C and W in the loss effect for the median language, and in how many of 27 languages C is larger. At INT4, processing matters most for single features and the new input for group loss.

weight term is larger. Attribution therefore depends on both the chosen edit and the outcome.

The same edit drifts at four bits. At legacy INT6, output-effect cosine is 0.984 and relative error is 0.165, about 1.88 INT16 reference floors. At INT4, cosine is 0.768 $[0.760, 0.776]$ and relative error 0.664. Lower relative error means the same edit gives a more similar output change. Matched RTN and GPTQ reduce this drift without removing it.

Control directions also drift substantially. A later comparison matched FP effect size on 144 FLORES pairs and 107 Wikipedia pairs. Real-direction INT4 error norms are 1.14 $[1.07, 1.25]$ and 1.18 $[1.05, 1.24]$ times the swapped-control norms. This shows a modest extra component on the selected subsets, alongside a large shared component.

Using control-direction sensitivity improves held-out legacy INT4 error prediction by 21.5% $[12.7, 29.4]$. At INT6 the gain is 10.9% $[-1.2, 21.4]$, so the original rule requiring utility at both widths is not met. A similar INT16 gain suggests sensitivity to perturbations generally. Activation survival and the scalar-loss gradient add no clear prediction gain for this vector-error target. The gradient does predict its own loss target moderately ($r = 0.63$ at FP, 0.58 for the INT4 shift); we did not compute a vector-gradient approximation.

Averaging effects across contexts makes preservation look stronger. For legacy INT4, cosine after averaging is higher by 0.126 than the average of context-level cosines.

Fresh text gives a similar INT4 pattern. On the six-language Wikipedia panel, INT4 cosine is 0.754, close to the FLORES value. Control-direction sensitivity improves INT4 prediction by 19%, and at INT6 it meets the utility rule.

10 Discussion and Conclusion

The central finding is a dissociation: activation survival and functional preservation come apart. At four bits, 87% of language features look "dead" by the standard activation cutoff, yet almost all still produce recognizable output effects. A direction-swap control (Figure 4b) shows that the similarity loss at INT4 is generic to downstream computation, not specific to the features themselves. Restoring the features' original activations recovers only 2.6% of the quantization damage, so the features are not where the harm sits. These three results together say that the features survive functionally but carry almost none of the model's loss from compression.

The selected groups contribute to next-token prediction in all 27 languages on both Gemma sizes, with dose–response confirmed in every language. The 9B confirmation met its informative frozen predictions, and the main patterns hold under all three quantizers. The weaker Llama result (21 of 27 languages) bounds the finding to the Gemma family.

Activation strength, not selectivity or resource tier, predicts which features stop firing, and feature counts follow how distinct a language's input is, not its resources. The account stops at reading comprehension, with no clear group effect.

Limitations, Challenges and Future Work

Most evidence comes from Gemma-2-2B at one layer and one fixed SAE. Not testing early and late layers, where language-specific neurons concentrate, or other dictionaries is a limitation of this work. Only one larger Gemma model and one smaller Llama setup were added, so results may differ with other models, layers or dictionaries. Features can mix or split concepts, and large edits can create unusual hidden states. Activity matching and direction controls reduce some differences without matching every feature property. The results apply to the specified edits and outcomes.

Most FLORES analyses reuse examined sentences. Holding out languages or features does not necessarily hold out documents. Intervals condition on the feature groups and other fixed choices. Language tiers also mix script, tokenization, training exposure and competence. Truncating inputs to a fixed token budget removes the length confound but may change internal representations. Every comparison is paired, with the same truncated tokens at each precision and for language and control groups, so truncation shifts both sides of each contrast equally and cannot by itself create a precision or language-versus-control difference. A direct test would add, for a subset of languages, complete sentences of equal token length and check whether excess harm and survival change, at the risk of a sentence-complexity confound. Belebele tests one comprehension task, while the external Wikipedia panel has six languages and shares a domain with GPTQ calibration.

The quantization experiment rounds weights during floating-point inference. It does not measure integer-kernel deployment or activation quantization. Calibration and error compensation are not varied independently. Small effects can depend on hardware and arithmetic precision. Additional independent text, dictionaries and precision comparisons would test broader generalization.

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A Language Panel and Reproduction

Tier	FLORES-200 language codes
High	eng_Latn, deu_Latn, spa_Latn, fra_Latn, por_Latn, rus_Cyrl, zho_Hans, jpn_Jpan, kor_Hang
Mid	hin_Deva, ben_Beng, tam_Taml, tur_Latn, vie_Latn, ind_Latn, tha_Thai, pol_Latn, ukr_Cyrl
Low	kan_Knda, tel_Telu, yor_Latn, sun_Latn, amh_Ethi, ibo_Latn, hau_Latn, khm_Khmr, lao_Laoo

Table 1: Fixed descriptive tiers, nine languages each.

The repository is https://github.com/TH4KUR/always-late; committed evidence maps identify result files, freezes and hashes. The artifact dataset is https://huggingface.co/datasets/karma-skz/always-late-artifacts; access is private and not all item-level files are public. Run summaries are at https://wandb.ai/karma-skz-/anlp-project. The current evidence map covers the whole completed study. Earlier report wording is archived verbatim.

The frozen environment records Python 3.10, PyTorch 2.14.1, TransformerLens 3.9.0, SAE Lens 6.54.5 and Transformers 5.19.0. Most 2B jobs used 11 GB RTX 2080 Ti accelerators; the supplied 9B run used a rented 96 GB RTX PRO 6000. Checks include sham identities, restored FP weights, quantized-weight hashes and analysis reproductions; new quantizer arms reproduce FP reference values exactly, and the activation-decomposition audit reproduces prior correlations within 3×10^{-8} and checks its identity to about 10^{-14}. The 9B review checks raw-file provenance and headers and reproduces tables byte-identically; it does not independently repeat the supplied GPU inference.

Bracketed ranges in the text are 95% bootstrap intervals unless stated otherwise. Resampling units follow the experiment: paired sentences for group loss, passages for Belebele, languages for F5 prediction and reviewed 9B comparisons, and features plus source sentences for transfer. Intervals hold other choices, such as groups and controls, fixed. They do not describe uncertainty over all model or SAE choices.

All numbers here derive from completed saved outputs; this revision introduces no model run, and the feature-count comparison is a descriptive summary of saved selections.

INT4 measurement	Legacy	Matched	GPTQ
Active-feature survival (%)	12.7	18.6	30.5
Mean loss increase (nats/token)	0.590	0.385	0.130
Median group-effect ratio	1.54	1.36	1.22
Median effect cosine (W_1)	0.839	0.875	0.923
Transfer cosine	0.768	0.822	0.877
Belebele accuracy (%)	23.6	29.1	31.8

Table 2: The main INT4 measurements of Gemma-2-2B under the three quantizers. Rows answer different questions. Survival averages active-feature shares per language; W_1 uses F5 features; transfer uses eligible feature–context pairs; Belebele uses 18 languages.

B Original Test Rules and Outcomes

Table 3 lists the original test families and outcomes. Its codes link the results to the analysis records. The final RQs organize the completed work without replacing these tests. A small positive estimate remains a result even when it misses a practical target.

C Additional Results

Figure 6 shows activation survival and selected-feature counts by resource tier. Table 2 gives the main INT4 measurements under the three quantizers; the qualitative pattern is the same.

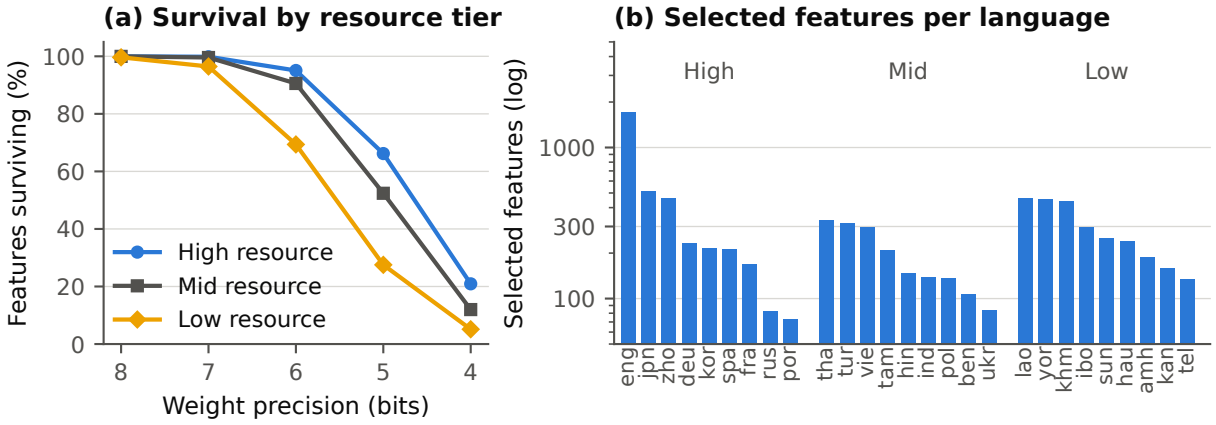


Figure 6: Resource tiers (Gemma-2-2B, legacy RTN; nine languages per tier). (a) Mean active-feature survival: low-resource languages lose more features at every width, but accounting for activation strength removes most of this gap (Section 6). (b) Number of selected language-selective features per language: counts do not follow resource tier.

Block	Rule or question	Completed outcome and status
Initial contrasts	Three selective-survival/tier contrasts; Holm within family	No rejection under original BOS-masked RTN. Matched/GPTQ reruns are exploratory additional families.
Association	FP establishment and peak retention	6,663/8,086 established; 70.8% retain peak at legacy INT4. Margin prediction is descriptive.
Functional F1–F4	Excess group loss, donor recovery, retained/lost subgroup effects	27/27 positive FP group effects; INT4 K64 recovery 2–3%; small subgroup gaps. Exploratory.
Belebele	Positive FP effect required before five precision statements	S0 not met; S1–S5 undefined. Original held-out family confirmatory; whole-group/prompt/positive-control checks exploratory.
Study G	H-G1/2 suppression dose/specificity; H-G3 boost dose; H-G6 consistency	Met under legacy RTN. Strict precision ordering H-G4 not met; a two-error tolerance sensitivity does not alter the original verdict.
Repair and gain	H-G5 generic/group-specific repair and inverse-shrinkage optimum	Main criteria not met. Fresh-sentence extension C meets its $1.1\times$ rule only; D shrinkage relation not supported.
State-defined G	Groups move; new groups more consequential; dose and repair	Group movement, dose and group-specific repair rules met; greater consequence and generic repair rules not met. Exploratory.
F5 and prediction	Identity, effect drift, association discrimination; continuous R_1	High identity retrieval despite drift. Binary association inconclusive/non-discriminative; continuous gain 3.1–4.4%, positive intervals, below 10% target.
Measurement audit	Gain/noise identity, scale separation, adjusted tiers and extra predictors	Identity verified; scale separation met by convention. Tier estimates mixed; gain/ν increments include zero. Exploratory.
Transfer	Fixed edit; H1 residual drift, H3 prediction, H4 averaging; group A3	Main H1/H4 supported; H3/A3 mixed; combined utility not met. External RTN utility met; GPTQ INT6 unresolved. Exploratory.
Llama	Frozen existence/dose/INT4 consistency rule	Not consistent (P1 21/27, 24 required); FP median effect positive. INT4 F5 association/prediction modestly positive. Exploratory.
Gemma-2-9B	FP effects, precision ladder, three-arm comparison	Positive 27/27, INT6 near FP, INT4 increase; all frozen predictions met. Two defective criteria documented.

Table 3: Completed study inventory. Outcomes follow each test's original rule. The main text and evidence map give estimates, uncertainty and explanations.